

COMS30035, Machine learning: k-means and mixtures of Gaussians

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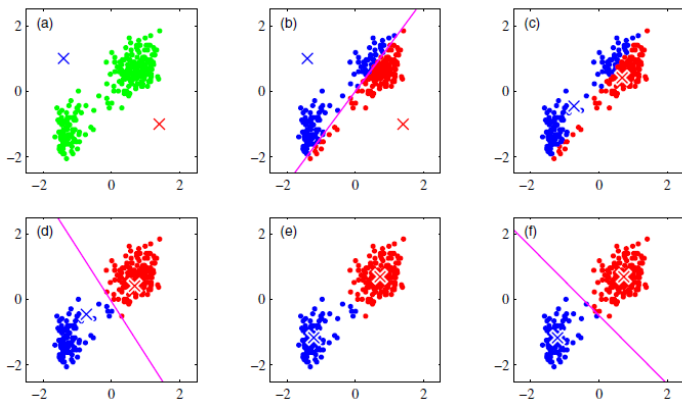
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Clustering

- ▶ Supervised vs Unsupervised learning.
- ▶ **Two methods:**
 - ▶ k -means: simple distance-based clustering.
 - ▶ Gaussian Mixture Models (GMMs): probabilistic soft clustering.

k -means for clustering



k -means optimisation

- ▶ $r_{nk} = 1$ if datapoint $\mathbf{x}_n$ is assigned to cluster k .
- ▶ $\boldsymbol{\mu}_k$ is the mean of cluster k .

$$r_{nk} = \begin{cases} 1 & \text{if } k = \arg \min_j \|\mathbf{x}_n - \boldsymbol{\mu}_j\|^2 \\ 0 & \text{otherwise} \end{cases} \quad (1)$$

$$\boldsymbol{\mu}_k = \frac{\sum_n r_{nk} \mathbf{x}_n}{\sum_n r_{nk}} \quad (2)$$

k -means objective function

- ▶ We want each point $\mathbf{x}_n$ to be close to its assigned cluster mean $\boldsymbol{\mu}_k$.
- ▶ Objective: minimize total squared distance to cluster centers.

$$J = \sum_{n=1}^N \sum_{k=1}^K r_{nk} \|\mathbf{x}_n - \boldsymbol{\mu}_k\|^2$$

- ▶ Optimize by alternating:
 1. Initialize K centers.
 2. Repeat until convergence (cluster assignments stop changing).
 - ▶ **E-step:** Assign each $\mathbf{x}_n$ to its nearest $\boldsymbol{\mu}_k$.
 - ▶ **M-step:** Update each $\boldsymbol{\mu}_k$ as the mean of its assigned points.

Pros and cons of k -means

Advantages

- ▶ Simple and fast: scales linearly with the number of data points N , clusters K , and dimensions D .
- ▶ Easy to implement and interpret — each cluster is summarized by its mean μ_k .
- ▶ Works well for roughly spherical and well-separated clusters.
- ▶ Often a good first choice for exploratory data analysis or initialization (e.g., for EM in GMMs).

Disadvantages

- ▶ Requires pre-specifying the number of clusters K .
- ▶ Assumes clusters are isotropic (spherical) and of similar size.
- ▶ Sensitive to initialization — can get stuck in local minima.
- ▶ Sensitive to outliers and noise (means can be pulled by extreme points).
- ▶ Produces *hard* assignments only — no uncertainty in cluster membership.

Gaussian mixture distribution

A Gaussian mixture model (GMM) assumes that data is generated from a mixture of K Gaussian components: [Bis06, §9.2]

$$p(\mathbf{x}) = \sum_{k=1}^K \pi_k \mathcal{N}(\mathbf{x} | \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k) \quad (3)$$

We can associate the *mixing coefficients* π_k with a K -dimensional latent variable $\mathbf{z}$ where:

$$z_k \in \{0, 1\}, \quad \sum_k z_k = 1, \quad p(z_k = 1) = \pi_k \quad (4)$$

Then, the marginal likelihood can be written as:

$$p(\mathbf{x}) = \sum_{\mathbf{z}} p(\mathbf{z}) p(\mathbf{x} | \mathbf{z}) = \sum_{k=1}^K \pi_k \mathcal{N}(\mathbf{x} | \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k) \quad (5)$$

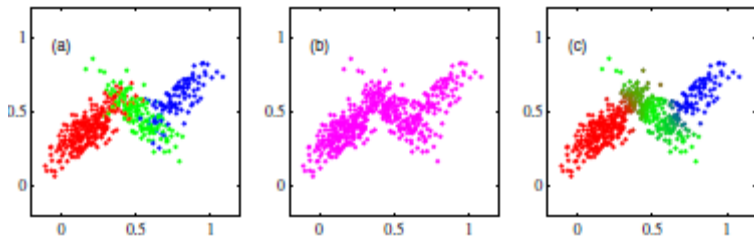
Responsibility and sampling

- ▶ Now we have a full joint distribution $p(\mathbf{x}, \mathbf{z})$ we can define the *responsibility* that component k has for ‘explaining’ observation $\mathbf{x}$

$$\gamma(z_k) = p(z_k = 1 | \mathbf{x}) \quad (6)$$

- ▶ To sample from a Gaussian mixture just use ancestral sampling: sample from $p(\mathbf{z})$, and then from $p(\mathbf{x} | \mathbf{z})$.

Soft clustering with Gaussian mixtures



More clustering with Gaussian mixtures

- ▶ If we want we can put restrictions on the covariance matrices of the Gaussians in the mixture.
 - ▶ Spherical: $\Sigma_k = \sigma_k^2 I$
 - ▶ Diagonal: $\Sigma_k = \text{diag}(\sigma_1^2, \sigma_2^2, \dots, \sigma_K^2)$
 - ▶ Full: No constraint; captures correlations between features.
- ▶ Let's have a look at [Mur22, p.729]

(Soft) clustering by MLE of a Gaussian mixture

- ▶ Given data $\mathbf{X}$ (and a fixed number K of component Gaussians) we can use MLE to estimate a particular Gaussian mixture distribution.
- ▶ Here's the log-likelihood [Bis06, 433]:

$$\ln p(\mathbf{X}|\boldsymbol{\pi}, \boldsymbol{\mu}, \boldsymbol{\Sigma}) = \sum_{n=1}^N \ln \left\{ \sum_{k=1}^K \pi_k \mathcal{N}(\mathbf{x}_n | \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k) \right\} \quad (7)$$

- ▶ Number of problems:
 1. Possible singularities
 2. Symmetry/nonidentifiability
 3. No closed form for the MLE
- ▶ Queue the EM algorithm ...

Comparing k -means and GMMs

- ▶ k -means: hard assignments, spherical clusters.
- ▶ GMM: soft assignments, elliptical clusters.
- ▶ GMM generalizes k -means when $\Sigma_k = \sigma^2 I$ and π_k are equal.

Reading

- ▶ Bishop §9.1 (you can skip §9.1.1 but it's an interesting read).
- ▶ Bishop §9.2 up to §9.2.1
- ▶ Murphy §21.3 up to §21.3.3

Problems and quizzes

- ▶ Bishop Exercise 9.1
- ▶ Bishop Exercise 9.3
- ▶ Quizzes:
 - ▶ Week 5: k-means and Mixtures of Gaussians



Christopher M. Bishop.

Pattern Recognition and Machine Learning.

Springer, 2006.



Kevin P. Murphy.

Probabilistic Machine Learning: An introduction.

MIT Press, 2022.